Carl Brown

Software Engineer · C#/.NET · Claude Code & Agent Engineering

Oklahoma, USA · carllbrown87@gmail.com · LinkedIn · GitHub

PROFESSIONAL SUMMARY

Software engineer with 12+ years of professional .NET experience and ownership of design and delivery across operational applications, payment services, and distributed infrastructure. Daily AI-assisted development experience with Claude Code and Codex, including building an agent harness used daily by teammates. Uses C#/ASP.NET Core, SQL Server, and Azure to deliver reliable operational software, modernize legacy systems, and improve application performance. Helps teammates deliver maintainable changes through architecture reviews, mentoring, and clear standards for validating AI-generated code.

AI ENGINEERING & SELECTED IMPACT

- **Built Mega Desk, an agent harness used daily by teammates:** integrates Claude Agent SDK and Codex CLI with Azure DevOps work-item prioritization, implementation handoff, PR reviews, comment resolution, and merge-conflict workflows.
- **Engineered agent execution and review workflows:** managed Git worktrees, branch-aware Claude session resumption, streamed and persisted run events, and human diff review before delivery actions. Added handling for concurrent resolution pushes and remote branch changes. https://conorbrown.orangecastle.net/projects/mega-desk.
- Authored **CLAUDE.md conventions** covering DDD, dependency boundaries, modern C#, and a reference vertical slice; reinforced standards through code reviews and workshops.
- Reduced CI/CD duration from roughly **40–60 minutes to 19 minutes**; reduced core test execution from 8–9 minutes to approximately 21 seconds through coverage-tooling changes.
- Built remote management infrastructure serving **2,000+ servers** and an Azure Service Bus usage-logging platform designed around a **99.9% delivery SLA**.

PROFESSIONAL EXPERIENCE

MIRATECH — April 2025–Present

Software Developer III · Architecture and platform leadership responsibilities

- Modernized a legacy .NET Framework MVC application to .NET 10; builds C#/ASP.NET Core Web APIs and React/TypeScript interfaces with SQL Server persistence.
- Uses Claude Code in implementation, refactoring, testing, and legacy-code workflows; checks proposed changes against domain rules, architecture boundaries, tests, and review before delivery.
- Guides engineers through design decisions and code reviews; led event-storming and DDD implementation sessions to make business rules and ownership explicit.
- Implemented DDD/CQRS workflow and rule-builder capabilities across aggregates, APIs, persistence, UI, and end-to-end testing, while guiding architectural reviews and workshops.
- Led a coordinated modernization path toward Azure App Service, including runtime configuration, Entra integration, Key Vault identity, parallel pipelines, staging validation, and rollback planning.
- Supported a product-line transition involving catalog identity, component modeling, pricing, drawings, indexing, and administrative tooling.
- Modernized CI/CD and test workflows, including Playwright-based regression coverage; codified engineering standards in a CLAUDE.md reference guide.

IXOPAY June 2021–April 2025
Senior Software Engineer

- Built payment-related microservices including a containerized Card Account Updater to address subscription churn caused by expiring payment credentials.
- Engineered asynchronous API usage telemetry with Azure Service Bus and a 99.9% delivery SLA for billing-related data.
- Introduced integration-test infrastructure and secure runtime configuration patterns across cloud services.

MissGroup / Hostek January 2014–June 2021
Software Engineer

- Rebuilt web-hosting management workflows with .NET Web APIs, React, and an orchestration/gateway layer.
- Designed remote infrastructure and deployment tooling operating across 2,000+ Windows servers, including IIS, firewall, certificate, and service management.
- Built secure credential management and remote access functionality supporting controlled sharing.

TECHNICAL BREADTH

Backend and architecture: C#, .NET Framework and .NET 10, ASP.NET Core / Web API, MVC, Entity Framework Core, DDD, CQRS, Clean Architecture, event-driven design, microservices and modular monoliths.
Cloud and platform: Azure App Service, Azure Service Bus, Key Vault, App Configuration, Docker, Azure DevOps, GitHub Actions, observability.
Product and data: Blazor WASM, React, TypeScript, React Native/Expo, SQL Server, PostgreSQL, MySQL, MongoDB, Cosmos DB, query/index tuning, Playwright, xUnit, NUnit, and integration testing.
AI development: Claude Code, Claude Agent SDK, Codex CLI, custom agent tooling, CLAUDE.md context conventions, agent workflows, prompt context, code review and validation.

INDEPENDENT PRODUCTS AND CERTIFICATION

Pathway: software engineering education platform · **Theoria:** interactive music education platform. Project walkthroughs: https://conorbrown.orangecastle.net/projects.
Certified Scrum Developer — Scrum Alliance